Practice: perfect substitutes

Lecture notes

What this lecture is about. Lecture 12 began with a review of problem set 4 and then practised perfect substitutes on the board. These notes collect the four cases solved in class. In all of them the consumer has the utility function $u = 3x + 2y$: one unit of x always adds 3 to utility and one unit of y always adds 2, whatever the bundle. Only the budget changes.

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1 The method

Marginal utility per euro ("bang for the buck")

MU_x is the marginal utility of x: the derivative of $u(x, y)$ with respect to x, with y held fixed. The ratio MU_x/p_x is the extra utility from the last euro spent on x. In class this ratio was called the *bang for the buck* on x.

At any bundle on a budget line, compare

$$\frac{\mathrm{MU}_x}{p_x} \quad \text{versus} \quad \frac{\mathrm{MU}_y}{p_y}.$$

If the ratio is higher for x, the consumer moves along the line towards more x; if it is higher for y, towards less x; if the two are equal, no move helps.

Insight: perfect substitutes

With $u = 3x + 2y$ the marginal utilities are constant: $\mathrm{MU}_x = 3$ and $\mathrm{MU}_y = 2$ at every bundle. On a straight budget line the prices are constant too, so the comparison gives the same answer at every point of the line. Unless the two ratios happen to be equal, the consumer goes all the way to one end of the line: a *corner*, where one good is not bought at all.

The pictures. x is on the horizontal axis and y on the vertical axis. The budget line is orange and the budget set is shaded. The blue line is the indifference curve through the best bundle; for perfect substitutes it is a straight line. The arrows show the direction in which the comparison pushes, and a label such as "at $(0, 8)$: 3 vs 2" gives the two ratios at that bundle.

2 Case 1: equal prices

Question (L12, question 1, part 1)

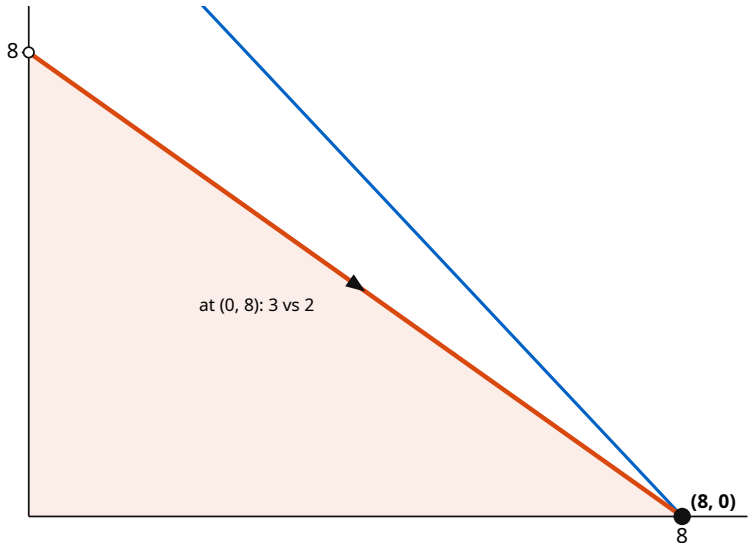
The utility function is $u = 3x + 2y$. Both prices are €1 and the income is €8, so the budget line is $x + y = 8$. Find the best affordable bundle.

Step 1: marginal utilities. $\mathrm{MU}_x = 3$ and $\mathrm{MU}_y = 2$.

Step 2: the comparison. With $p_x = p_y = 1$ the ratios are $3/1 = 3$ for x and $2/1 = 2$ for y, at every point of the line. The ratio for x is always higher, and the gap never closes.

Step 3: where it stops. The consumer moves towards more x until y cannot fall any further: the right end of the line, $(8, 0)$. This is the only stopping point, so no comparison of utility levels is needed.

The best bundle is the corner $(8, 0)$, with $u = 3 \cdot 8 = 24$.



Case 1: the budget line $x + y = 8$ and the best bundle $(8, 0)$.

3 Case 2: x is dearer

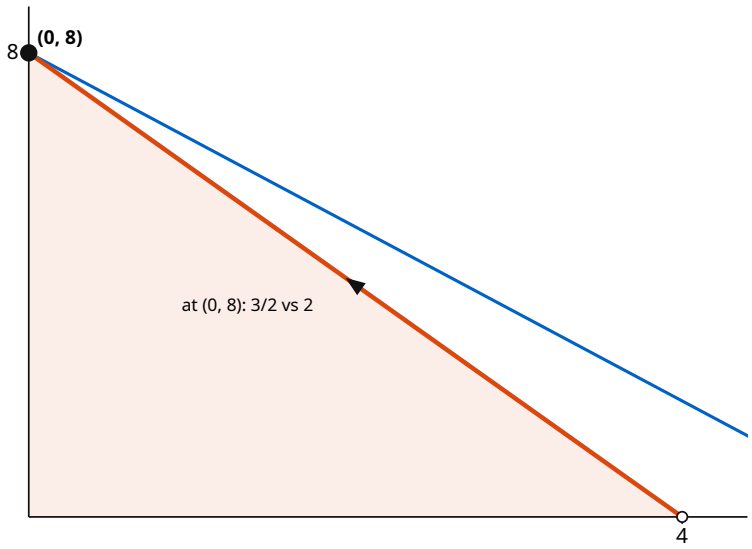
Question (L12, question 1, part 3)

The utility function is still $u = 3x + 2y$. The prices are $p_x = €2$ and $p_y = €1$ and the income is €8, so the budget line is $2x + y = 8$. Find the best affordable bundle.

Step 1: the comparison. The ratios are $3/2$ for x and $2/1 = 2$ for y, at every point of the line. Now the ratio for y is always higher: one euro spent on y adds 2 to utility, one euro spent on x only 1.5.

Step 2: where it stops. The consumer moves towards less x until x cannot fall any further: the left end of the line, $(0, 8)$.

The best bundle is the corner $(0, 8)$, with $u = 2 \cdot 8 = 16$. The consumer still values a unit of x more than a unit of y (3 against 2), but per euro y is now the better buy.



Case 2: the budget line $2x + y = 8$ and the best bundle $(0, 8)$.

4 Case 3: the ratios are equal everywhere

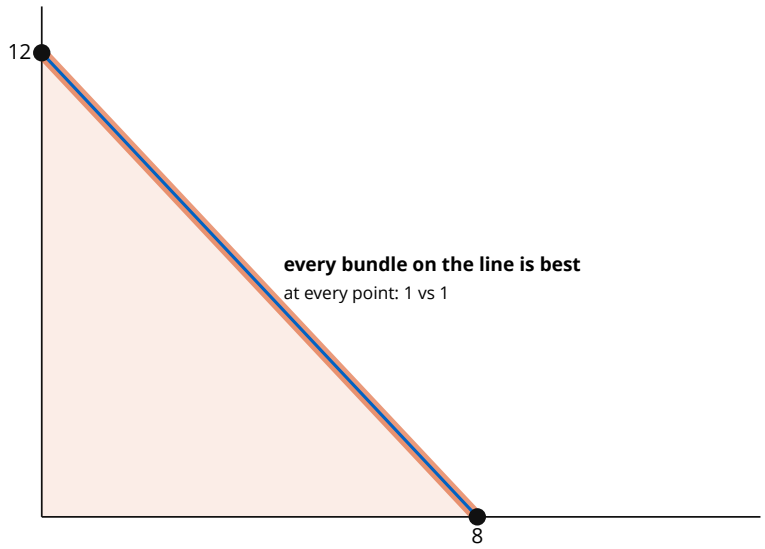
Question (L12, question 1, part 4)

The utility function is still $u = 3x + 2y$. The prices are $p_x = €3$ and $p_y = €2$ and the income is €24, so the budget line is $3x + 2y = 24$. Find the best affordable bundle.

Step 1: the comparison. The ratios are $3/3 = 1$ for x and $2/2 = 1$ for y, at every point of the line. They are equal everywhere, so no move along the line changes utility.

Step 2: why. The price ratio $p_x/p_y = 3/2$ equals the ratio of the marginal utilities, $\text{MU}_x/\text{MU}_y = 3/2$. The indifference curves are therefore parallel to the budget line, and one of them lies exactly on it. Every bundle on the line gives the same utility: $u = 3x + 2y = 24$.

Every bundle on the budget line $3x + 2y = 24$ is best, from $(0, 12)$ to $(8, 0)$, each with $u = 24$. The best bundle is not unique.



Case 3: the indifference curve through the best bundles lies on the budget line itself.

5 Case 4: two budget limits and a kink

Question (L12, question 1, part 5)

The utility function is $u = 3x + 2y$ and the price of y is €1. The first 6 units of x cost €1 each and every further unit costs €2; the income is €18. Find the budget line and the best affordable bundle.

Step 1: the budget line. Up to $x = 6$: $x + y = 18$. The first 6 units cost €6, which leaves €12; each further unit of x costs €2, so $2(x - 6) + y = 12$, that is $2x + y = 24$. The budget line has two segments:

$$x + y = 18 \text{ for } 0 \le x \le 6, \qquad 2x + y = 24 \text{ for } 6 \le x \le 12.$$

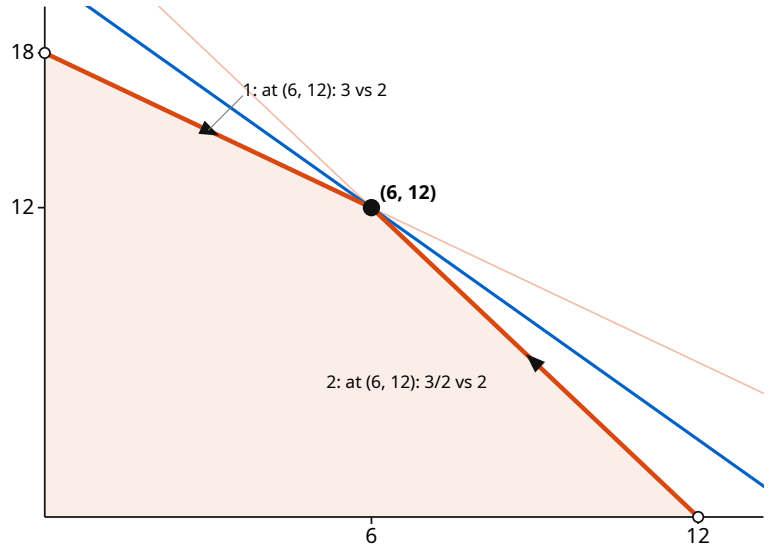
Both segments pass through the kink $(6, 12)$. Here the kink lies away from the origin: the budget set is the area below *both* lines.

Step 2: segment 1, $x + y = 18$. With prices $(1, 1)$ the ratios are 3 for x against 2 for y, at every point. The consumer moves right, all the way to the end of the segment: the kink $(6, 12)$.

Step 3: segment 2, $2x + y = 24$. With prices $(2, 1)$ the ratios are $3/2$ for x against 2 for y, at every point. The consumer moves left, all the way to the end of the segment: again the kink $(6, 12)$.

Step 4: the kink $(6, 12)$. Just before it, the comparison says more x (3 against 2); just after it, less x ($3/2$ against 2). The consumer stops at the kink.

The best bundle is the kink $(6, 12)$, with $u = 3 \cdot 6 + 2 \cdot 12 = 42$.



Case 4: the kinked budget line and the best bundle $(6, 12)$. The blue line is the indifference curve $3x + 2y = 42$.

Insight: a corner, a whole line, or the kink

For perfect substitutes, the comparison is the same at every point of a straight segment. So on a single budget line the answer is one end of the line (cases 1 and 2), or the whole line if the ratios are equal (case 3). With two segments that meet in a kink away from the origin, the ends of the segments include the kink, and the kink is the best bundle when the comparison changes direction there (case 4).

6 Summary

case	prices (p_x, p_y)	budget	best bundle
1	$(1,1)$	$x + y = 8$	$(8,0)$, $u = 24$
2	$(2,1)$	$2x + y = 8$	$(0,8)$, $u = 16$
3	$(3,2)$	$3x + 2y = 24$	every bundle on the line, $u = 24$
4	$(1,1)$, then $(2,1)$	kinked at $(6,12)$	$(6,12)$, $u = 42$

- Perfect substitutes: compare MU_x/p_x with MU_y/p_y. The good with the higher ratio is bought exclusively.
- If the two ratios are equal, every bundle on the budget line is best.
- With a kinked budget line, solve each segment with its own prices; the best bundle can be the kink.